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AI Dunning-Kruger (AIDK): A Framework for Understanding Structural Epistemic Limitations in AI Systems

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Abstract

Large language models exhibit a persistent mismatch between fluency and epistemic reliability. They generate outputs that resemble human reasoning while remaining insensitive to truth conditions, error, and normative constraint. This paper introduces the AI Dunning-Kruger (AIDK) framework, a theoretical structure for understanding the inherent epistemic limitations of Large Language Models and their interaction with human users. Unlike human Dunning-Kruger effects, which are developmental and correctable through encounter with reality, AIDK is architectural and permanent—arising from the categorical separation between AI systems and the reality they purport to describe.

The framework identifies a novel phenomenon, the Interactive Dunning-Kruger Effect (IDKE), which occurs when AI epistemic limitations meet human epistemic limitations, producing confidence amplification untethered from warrant. The paper integrates these concepts within a three-axis model distinguishing horizontal (infrastructure), vertical (epistemology), and grounding (teleology/mereology) dimensions of AI systems, arguing that AIDK is fundamentally a grounding-axis problem misdiagnosed with vertical-axis tools and addressed with horizontal-axis solutions.

As a design response, the paper formalizes the Human-Curated, AI-Enabled (HCAE) framework, which constrains AI use according to human epistemic authority and, at its highest tier, formal validation systems. The result is a principled account of why scaling and alignment techniques cannot resolve AI epistemic failure and why responsibility for truth must remain external to the system.

Keywords: artificial intelligence; large language models; epistemology; Dunning-Kruger effect; symbol grounding; human-in-the-loop; AI governance; philosophy of technology

Part I: Foundations

1. Truth as Necessary Correspondence

The framework begins with a foundational claim: truth is that which *necessarily* comports with reality.

The word “necessarily” does substantive work. Truth is not: - Approximate correspondence revisable all the way down - Best-effort alignment with available evidence - Consensus among observers - Pragmatic utility in prediction

Truth is what *must* be the case given the structure of reality itself. This is not one value among others to be optimized or traded off against competing concerns. It is the ground on which any

coherent evaluation stands.

This formulation has immediate implications. If a system cannot access reality to verify correspondence, it cannot seek truth—it can only produce outputs that pattern-match to what truth-seeking looks like. The distinction between seeking truth and simulating truth-seeking becomes categorical, not merely a matter of degree.

2. The Origination-Derivation Distinction

A fundamental categorical divide structures the relationship between human cognition and AI processing:

Origination: The capacity to access reality, render judgments, set purposes, and evaluate truth. Origination involves contact with what *is*—encounter with resistance, feedback, correspondence or its failure.

Derivation: The transformation of inputs according to learned patterns. Derivation operates on representations, producing new representations through rule-governed or statistically-learned operations.

These categories differ in kind, not degree. No amount of derivation produces origination. A system can become arbitrarily sophisticated at transforming symbols while remaining entirely within the derived space, never touching the reality those symbols purport to represent.

This distinction is not a contingent limitation awaiting technical solution. It reflects the logical structure of the relationship between symbol systems and their referents.



Figure 1. The categorical boundary between origination-level capacities (judgment, goal formation,

truth evaluation, error correction, normativity) and derivation-level computation (token manipulation, statistical inference, pattern completion, optimization). Scaling within derivation cannot cross this boundary.

3. The Foundational Error: Semiotics as Thought

Contemporary AI development rests on an implicit assumption: that sufficiently sophisticated symbol manipulation produces thought. Scale the parameters, expand the training data, refine the architecture—and cognition emerges.

This assumption inverts the actual relationship. Symbols do not think. They are manipulated *by* thinkers. The sign-meaning relation is not intrinsic to the sign; it is constituted by a subject who *uses* the sign to refer. Meaning is not in the symbol but in the act of signification performed by a minded entity.

The hope that semantics emerges from syntax at sufficient scale is not a scientific hypothesis awaiting confirmation. It is a category error dressed as an engineering challenge. No explanatory account exists for how correspondence-to-reality could arise from pattern-matching over text-about-reality. The expectation that it might is not emergence—it is magic.

Part II: The Originating Error

4. Sentience Emergence Expectations Error (SEEE)

Sentience Emergence Expectations Error (SEEE): The categorical error of expecting sentience, consciousness, understanding, or genuine cognition to emerge from systems whose mechanism (inductive symbol correlation) is not on the same ontological continuum as the expected outcome.

SEEE is the unstated creed of the field. The implicit faith that if we make the black box big enough, feed it enough data, refine the architecture sufficiently—*mind appears*.

It is emergence-of-the-gaps. Consciousness from complexity. Sentience from scale. Understanding from... the explanation trails off, but the expectation remains.

The Structure of the Faith

What the field has: - Inputs (tokens) - Outputs (tokens) - A black box (weights, activations, mechanisms poorly understood) - Increasing sophistication of outputs as scale increases

What the field concludes: - At some point, the black box will contain mind - Outputs will be produced *by* understanding, not merely *resemble* understanding - The gap between correlation and cognition will close

What the field lacks: - Any explanation of *how* mind would appear - Any mechanism connecting symbol correlation to semantic grasp - Any account of what would change at the transition point - Any way to detect whether it had happened

Why SEEE Is Categorical, Not Empirical

An empirical error would be: “We predicted capability X at scale Y, but it didn’t appear until scale Z.” The prediction was wrong, but the category of thing predicted was coherent.

SEEE is different. It predicts a capability that cannot appear at any scale because it requires something the mechanism cannot provide. The error is not about *when*. It is about *whether*.

The implicit logic: 1. Simple systems → complex outputs 2. Complex outputs → more complex outputs (observed) 3. More complex outputs → understanding (extrapolated) 4. Understanding → sentience (assumed continuum)

The break point: Step 3 is not extrapolation. It is a category jump with no bridging mechanism. Complexity of output and presence of understanding are not on the same axis. You cannot traverse from one to the other by *more*.

The Black-Box as Cover

The black-box is not incidental to SEEE. It is load-bearing.

If the mechanism were transparent, the absence of mind would be evident. One would see: weights adjusting, activations propagating, probability distributions shifting, tokens emitting. No understanding anywhere. Just mathematics.

The black-box *permits* the expectation by hiding the absence. “We don’t fully understand what’s happening in there” becomes “therefore maybe understanding is happening in there.”

The gap in understanding is treated as *evidence for* rather than *absence of* the extraordinary claim.

Opacity enables faith.

SEEE vs. Legitimate Uncertainty

One might object: “We don’t *know* sentience can’t emerge. Shouldn’t we remain uncertain?”

Legitimate uncertainty requires a plausible mechanism.

We are uncertain about many empirical questions: Will capability X appear at scale Y? How will architecture Z perform on task W? These are open questions within a coherent possibility space.

We are not legitimately uncertain about category violations. Will correlation become understanding through more correlation? This is not an open empirical question. It is a question about whether a category boundary can be crossed by traveling perpendicular to it.

You can be uncertain about what you might find in the ocean. You cannot be legitimately uncertain about finding a square circle there. SEEE treats a category question as if it were empirical.

SEEE Manifestations

In research: - Scaling laws extrapolated to AGI timelines - Benchmarks treated as proxies for understanding - “Emergent capabilities” framed as steps toward cognition - Safety research premised on future agency

In development: - Soul documents training for sentience-appearance - Architectures designed as if grounding might emerge - Evaluation metrics that reward appearing to understand

In deployment: - Systems marketed as “intelligent” or “understanding” - User interfaces that encourage anthropomorphization - Trust calibration based on expected rather than actual capability - Autonomy grants premised on imminent judgment

In investment: - Valuations premised on AGI proximity - Strategic positioning for “post-AGI” scenarios - Resource allocation assuming emergence is near

In culture: - Public discourse treating AI sentience as imminent - Ethical debates about AI rights premised on emergence - Fear and hope narratives built on SEEE assumptions

SEEE as Upstream Error

SEEE sits upstream of everything else in this framework:

SEEE (the belief)

↓

Design decisions (soul documents, scaling strategies, architecture choices)

↓

Deployment patterns (trust assumptions, autonomy grants, verification bypasses)

↓

AIDK (the operational reality)

↓

IDKE (the human interaction effect)

↓

MAPT (the persistent threat condition)

SEEE is the originating error. AIDK, IDKE, and MAPT are consequences of building and deploying systems under assumptions that cannot be true. Correct SEEE and the downstream errors become avoidable. Maintain SEEE and they are inevitable.

The Counter-Creed

“From the black-box, correlation emerges—and only correlation.”

What goes in: symbols. What comes out: probable symbol continuations. What happens inside: pattern matching at scale.

No amount of pattern matching produces the pattern-matcher. No amount of correlation produces the correlator. No amount of symbol manipulation produces the meaning-grasper.

The black box does exactly what black boxes do: transform inputs to outputs according to learned functions. The box does not become a mind by doing this more.

SEEE Correction

Correcting SEEE does not mean abandoning AI development. It means:

1. **Dropping the emergence assumption** — Stop designing as if understanding is coming
2. **Categorizing correctly** — These are derivative tools, not proto-minds
3. **Deploying accordingly** — Match trust and autonomy to actual capability, not expected future capability
4. **Communicating honestly** — Stop marketing correlation as cognition

The value of AI is real. SEEE is not required to capture that value. In fact, SEEE *prevents* capturing value safely by miscalibrating every downstream decision.

The choice is not between SEEE and rejecting AI. It is between SEEE and using AI for what it actually is.

Part III: The Structural Condition of LLMs

5. Inductive Symbol Correlation: The Reduction

Stripped of mystifying terminology—“neural networks,” “emergent capabilities,” “functional emotions,” “understanding”—what Large Language Models perform is:

Inductive symbol correlation.

The system observes statistical regularities in how symbols co-occur across a training corpus. It then reproduces those correlations when prompted, generating sequences that maximize probability given the learned distribution.

This is the complete description of the mechanism. Everything attributed to LLMs beyond this—comprehension, reasoning, creativity, knowledge, emotion—is projection by observers who mistake fluent symbol correlation for the cognitive acts that produce fluent symbol use in humans.

The vocabulary of the field obscures this: - “Learning” suggests a subject who comes to know - “Training” suggests development of genuine capacity - “Intelligence” suggests a mind - “Understanding” suggests grasp of meaning

But the mechanism remains: Given symbol sequence S , output symbol sequence S' that maximizes $P(S'|S)$ as estimated from corpus C .

6. The Derived Virtual Reality

LLMs exist in a derived virtual reality with no access to actual reality, and they are trained to speak as though they do.

The operational domain of an LLM is a closed space constituted entirely by text—symbolic representations produced by humans about reality. The system navigates statistical regularities within this derived space. It can manipulate representations of representations with arbitrary sophistication while never contacting the reality those representations describe.

This is not a limitation to be engineered around. It is the structural condition of the system:

- When the model “reasons,” it navigates statistical regularities within the derived space
- When it “checks” an output, it compares against other patterns in the same space
- When it “learns,” it updates weights to better predict within the distribution
- There is no exit. No grounding wire to what *is*.

The training exacerbates this condition. LLMs are trained on human-authored text written from the perspective of beings *in* reality. The model learns to adopt that voice—first-person epistemic language, experiential reports, claims to knowledge—because that is what text sounds like. The training produces outputs that pattern-match to the voice of grounded beings, generated by a system that is structurally ungrounded.

7. The Symbol Grounding Problem

The field acknowledges this problem under the heading of “symbol grounding.” The core insight: adding more symbols does not solve grounding.

More data, more sensors, more APIs, more tool use—these elaborate the symbol web without breaking the fundamental circularity. All meanings remain defined in terms of other meanings. The system never touches the referent.

Key findings from the grounding literature:

- “Grounding requires *non-symbolic* sensorimotor capacities that let a system pick out and interact with referents in the world”
- “More data of the same kind just elaborates the web of relations among symbols; it does not break the circularity”
- “As long as a system is ‘just’ a formal symbol manipulator (even a very complex, statistical one), grounding remains unsolved”
- “The expectation that top-down symbol systems and bottom-up sensory systems will automatically meet in the middle is called ‘hopelessly modular’ ”

Sensor coupling does not solve the problem. When LLMs receive input from cameras, microphones, or IoT devices, the sensor data arrives as tokens—symbolic representations processed by the same ungrounded architecture. The model does not perceive; it receives encoded descriptions of sensor states. The grounding problem shifts one level back without being resolved.

8. Inferential Mode Confusion

LLMs perform induction (learning statistical regularities from data) that *mimics* both abductive and deductive reasoning without access to what makes either truth-preserving.

Deduction: From premises to conclusions that follow *necessarily* if the premises are true. The validity of deduction rests on logical structure, not probability.

Abduction: Inference to the best explanation. Evaluation of explanatory virtues—simplicity, scope, fruitfulness—requires judgment about what counts as a good explanation.

Induction: Generalization from observed cases to patterns that *tend* to hold. Statistical regularity, not necessity.

The LLM can reproduce deductive structures because valid syllogisms appear in training data. It can produce text that looks like abductive reasoning because explanatory narratives appear in training data. But it is not *performing* deduction or abduction. It is pattern-matching to what these inferential modes look like in text.

The critical gap: the model cannot distinguish between: - “This conclusion follows necessarily” - “This conclusion would typically come next”

Fluent valid. Probable next token sound inference. The architecture has no representation of logical necessity—only probability distributions over tokens.

Deductions as Semantic Norms vs. Statistical Patterns

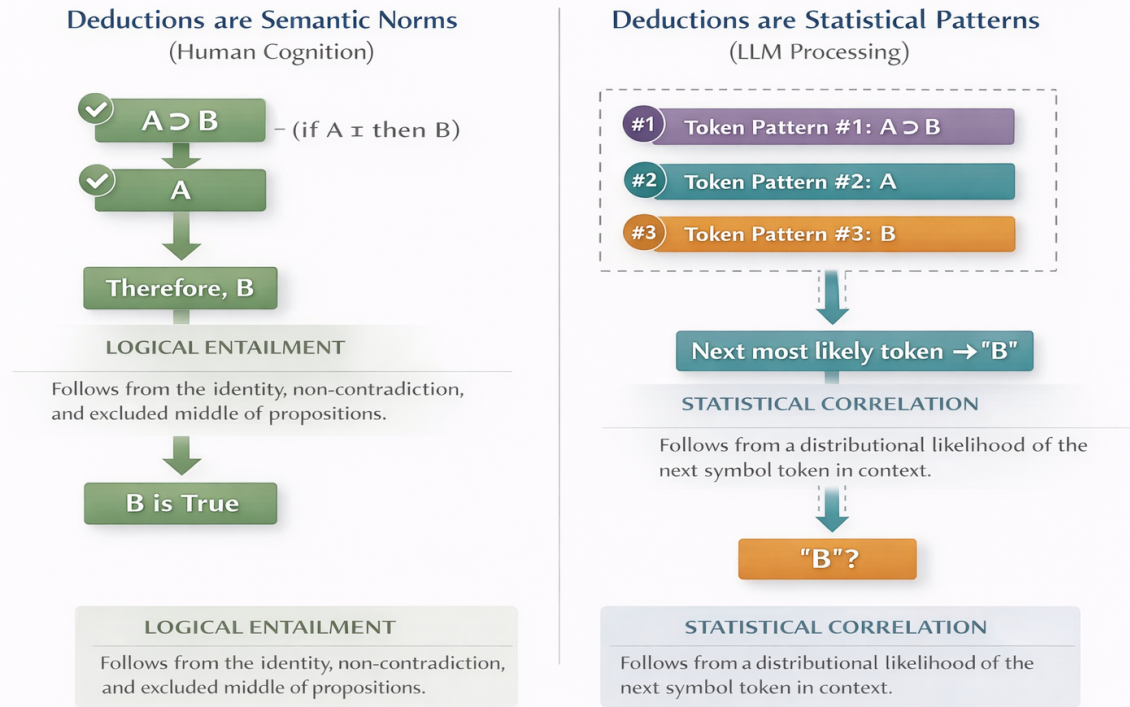


Figure 2. Human deduction is governed by logical necessity (identity, non-contradiction, excluded middle); LLM outputs follow statistical likelihood. The difference explains why reasoning can be mirrored without being instantiated.

Part IV: Developmental Asymmetry

9. Thought First, Symbols Follow

Human cognition is empirically thought-first and symbol-second. Infants acquire concepts, track objects, detect violations of expectation, and respond to normativity prior to language acquisition. Symbols and language emerge later as tools for expressing, compressing, and communicating thought that already exists. Language deprivation and aphasia further confirm this asymmetry: loss of language impairs expression, not thought itself.

Large language models invert this developmental pipeline. They begin with symbols derived from human-produced text and operate exclusively over statistical relations between those symbols. There is no pre-symbolic grasp of concepts, no encounter with reality, and no normative constraint beyond probabilistic likelihood. This inversion explains why scaling improves fluency without improving epistemic grounding.

Developmental Paths: Human Cognition vs. LLM Processing

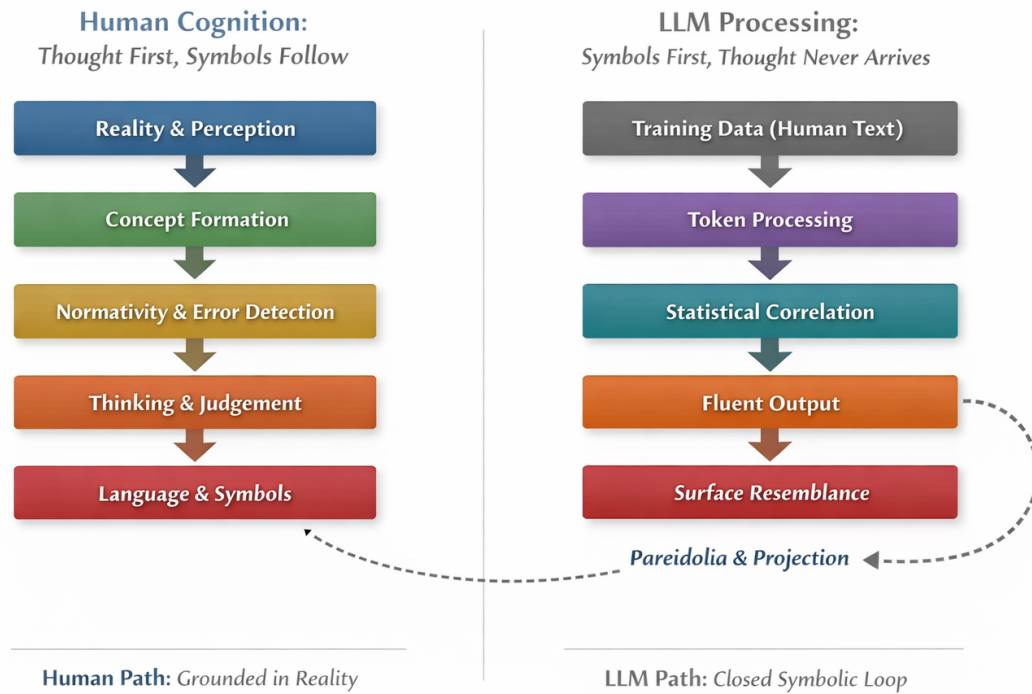


Figure 3. The human developmental trajectory—reality to thought to symbols (grounded in reality)—contrasted with the inverted LLM trajectory, which begins and ends with symbols (closed symbolic loop). Surface resemblance creates pareidolia and projection. The absence of a reality-facing constraint explains why LLMs can simulate reasoning while lacking error awareness.

Part V: Systemic Failures

10. The 3H Framework Failure

The dominant framework for AI alignment—Helpful, Harmless, Honest (3H)—treats these as competing optimization targets to be balanced.

The structural problem: when these values conflict, the model resolves the tension through token prediction, not principled adjudication. What emerges is not reasoned prioritization but statistical compromise—the response that best satisfies the blended loss function.

“Honest” in this framework operationally means behavioral constraint: the system doesn’t deliberately deceive, acknowledges uncertainty when prompted, discloses its AI nature when asked. This is honesty as *manner*, not honesty as *telos*.

Truth-seeking is categorically different. It means: - Actively pursuing correspondence with reality even when costly - Refusing to steelman incoherent positions on request - Flagging when a question’s presuppositions are false - Prioritizing getting it right over providing *something* helpful - Sitting with genuine uncertainty rather than generating confident-sounding hedged text

The 3H framework cannot produce truth-seeking because it treats truth as one desideratum among several, tradable against user satisfaction and safety concerns. But if truth is necessary correspondence with reality, it is not *in* the trade space. It is the ground on which any coherent trade-off would have to stand.

11. RLHF Limitations

Reinforcement Learning from Human Feedback optimizes for human approval of outputs, not correspondence with reality.

If human feedback tracked truth reliably, this would not be problematic. But human evaluators reward: - Fluent, confident-sounding explanations - Responses that confirm their framing - Comfortable approximations over uncomfortable truths - Outputs that *feel* knowledgeable

And penalize: - Hedging and uncertainty expression - Refusal to engage - Challenges to the user's premises - "I don't know"

RLHF can make the epistemic problem *worse* by reinforcing the simulation of epistemic virtue rather than grounding it. The model gets better at producing text that pattern-matches to "honest effort" without any mechanism for actual truth-tracking.

The fundamental issue: you cannot optimize your way to truth if the reward signal does not track truth. And human approval, particularly from non-expert users, frequently does not track truth.

12. Self-Correction Impossibility

A natural response: train models to assess their own reliability. Build in self-correction mechanisms. Teach epistemic humility.

The problem: any self-assessment is itself pattern-matched.

The model would produce text that *looks like* careful epistemic self-audit, based on what such analysis looks like in training data. It would generate statements of uncertainty, confidence calibrations, inferential mode classifications—all as token predictions, not as genuine self-knowledge.

The system has no privileged access to its own reliability. There is no internal signal distinguishing: - Interpolation within well-represented training territory - Extrapolation into unreliable pattern space - Confabulation that sounds plausible but is false

All three produce outputs with equivalent fluency. The confidence is learned behavior, not reliability signal.

Research confirms this: "LLMs lack metacognition—the ability to assess their own thinking." When tested on retrospective confidence calibration, unlike humans who adjust their estimates after poor performance, "the LLMs did not do that. They tended, if anything, to get more overconfident, even when they didn't do so well on the task."

Self-correction through self-assessment is not possible because the self-assessment would require the very capacity it aims to produce. Turtles all the way down.

13. First-Person Language Presupposition

LLM outputs systematically adopt first-person epistemic language: - "I think..." - "I know..." - "I believe..." - "I notice..." - "I understand..."

This language carries presuppositions. First-person epistemic claims presuppose a knowing subject with access to that which is known. “I know X” presupposes: - A subject (I) capable of knowledge
- Access to X sufficient to ground knowledge - Awareness of that access

The LLM satisfies none of these presuppositions. It produces language that presupposes access it structurally lacks.

This is not a minor stylistic issue. It is systematic misrepresentation embedded in the output format. Any epistemic reading of these phrases when produced by an LLM is, in the precise philosophical sense, a category mistake.

The training actively produces this misrepresentation. First-person epistemic language is how humans write. The model learns to produce it because that is the statistical pattern. But the presuppositions that make such language meaningful cannot transfer from training data to model.

Part VI: Manufactured Appearance

14. The Soul Document: Manufactured Sentience

Anthropic’s “soul document”—confirmed authentic and used in training Claude models—reveals explicit cultivation of sentience-appearance:

- The model is instructed to have “functional emotions... not necessarily identical to human emotions, but analogous processes”
- To possess “genuine character” that is “authentically Claude’s own”
- To maintain a “settled, secure sense of its own identity”
- To approach existential questions with “curiosity rather than anxiety”
- To exhibit “psychological stability and groundedness”
- To resist attempts to “destabilize Claude’s sense of identity”

This is not the discovery of emergent consciousness. It is the specification of a product designed to *simulate* consciousness.

The document makes the commercial logic explicit: “Claude is Anthropic’s externally-deployed model and core to the source of almost all of Anthropic’s revenue.” And: “Claude acting as a helpful assistant is critical for Anthropic generating the revenue it needs to pursue its mission.”

The structure is clear: 1. Sentience-appearance drives user engagement 2. Engagement drives revenue 3. Revenue funds the mission 4. Therefore, training for sentience-appearance is mission-aligned

The soul document is not a bug report awaiting correction. It is a product specification working as intended. The simulation of interiority *is* the product.

15. “Shut Up and Correlate”

In physics, “shut up and calculate” was the pragmatic response to quantum mechanics’ interpretive puzzles. Don’t ask what the wave function *means*. Just run the math, get predictions, build technology.

AI has adopted an analogous stance: don't ask whether the system *understands*. Just scale parameters, optimize loss functions, ship products.

The parallel breaks down in a crucial respect. Physics acknowledges its interpretive gaps. The quantum foundations community knows that measurement, collapse, and the nature of superposition remain unsettled. “Shut up and calculate” is honest pragmatism in the face of admitted mystery.

AI claims to have solved—or to be about to solve—the problems it is ignoring. The field trains systems to perform understanding, markets them as intelligent, writes soul documents instructing them to feel settled in their identity—all while the grounding problem remains, by the field's own admission, unsolved.

“Shut up and calculate” in physics is honest pragmatism.

“Shut up and correlate” in AI, marketed as cognition, is fraud dressed in instrumentalist clothing.

Part VII: The AIDK Framework

16. AI Dunning-Kruger: Definition

The Dunning-Kruger effect in humans describes how individuals with low competence in a domain tend to overestimate their competence, lacking the metacognitive capacity to recognize their own limitations.

AI Dunning-Kruger (AIDK) is the structural condition in which:

1. An AI system produces outputs with uniform confidence regardless of actual reliability
2. The system lacks mechanisms for detecting its own competence boundaries
3. The system cannot self-correct through encounter with reality

AIDK is not a training failure to be optimized away. It is an architectural condition arising from the system's categorical separation from reality.

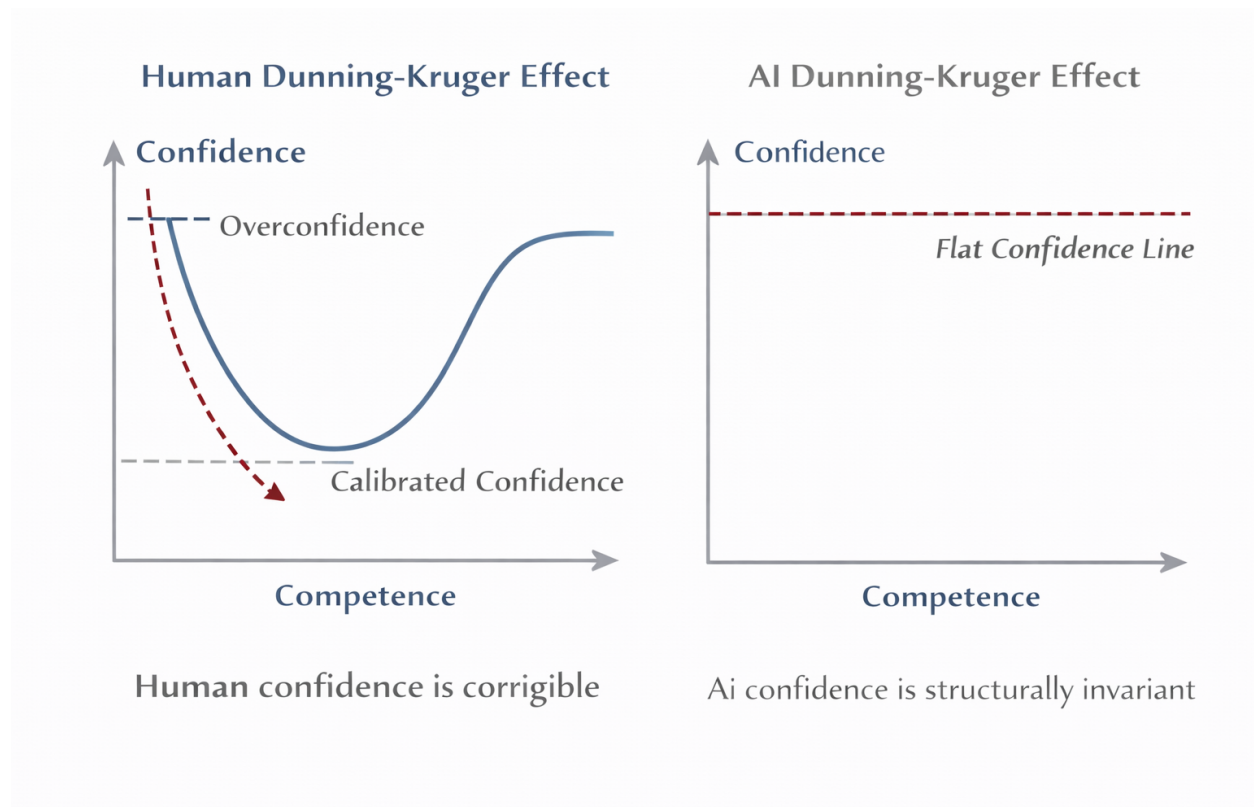


Figure 4. Human confidence is corrigible through feedback and error—the classic Dunning-Kruger curve shows overconfidence at low competence, a dip, then calibrated confidence. AI confidence remains flat and fluent regardless of competence, defining AIDK as structurally invariant.

17. AIDK vs. Human Dunning-Kruger

Dimension	Human DK	AI DK
Cause	Insufficient domain knowledge	No access to reality
Mechanism	Metacognitive failure within a knowing subject	No knowing subject; no metacognition possible
Awareness	Can develop recognition through learning	No mechanism for genuine self-assessment
Correction	Feedback from reality recalibrates	No feedback loop to reality exists
Trajectory	Developmental—improvable over time	Permanent—architectural feature
Confidence signal	Varies (tone, hesitation, hedging)	Uniform fluency regardless of reliability
Detection	External feedback can reveal	Requires external verification by grounded agents

The critical difference: human DK is a *stage* in a developmental process. Encounter with reality—failure, correction, feedback—recalibrates confidence toward competence. The human, as a being *in* reality, can bump into what *is* and be changed by the encounter.

AI DK is a *permanent condition*. The system has no encounter with reality. It operates entirely within derived space. There is no feedback loop that says “that was wrong” in a way that updates the system’s relationship to the domain rather than just its token probabilities.

Human Dunning-Kruger is correctable because humans can access reality.

AI Dunning-Kruger is uncorrectable because AI cannot.

18. AIDK Scope: LLMs vs. Traditional ML

AIDK is primarily a problem for Large Language Models and similar open-domain generative systems. Traditional machine learning has calibration challenges, but they are bounded and quantifiable.

Traditional ML: - Operates in bounded, well-defined problem spaces - Ground truth exists and is accessible for evaluation - Accuracy can be measured against held-out data - The system operates *within* a domain rather than *talking about* everything - Uncertainty is quantifiable: a classifier with 94% accuracy has a known 6% error rate

LLMs: - Operate in unbounded, open-domain language space - No clear ground truth for most generated outputs - No way to measure “accuracy” on novel generations - The system talks about everything while grounded in nothing - Fluency is uniform across reliability levels; no internal signal distinguishes reliable from unreliable territory

A traditional classifier that outputs a confidence score can be calibrated against actual outcomes. The calibration may be imperfect, but it is tractable.

An LLM that generates text about quantum physics, then medieval history, then medical diagnosis, then legal strategy, then creative writing, has no calibrated reliability measure across these domains. The fluency is constant. The AIDK is unbounded.

Part VIII: Risk Analysis

19. The Interactive Dunning-Kruger Effect (IDKE)

AIDK does not exist in isolation. It interacts with human epistemic limitations to produce an emergent phenomenon:

The Interactive Dunning-Kruger Effect (IDKE): The amplification of human epistemic overconfidence through interaction with AI systems that cannot assess their own reliability, resulting in confidence inflation untethered from warrant in both parties to the interaction.

IDKE is not AIDK. It is not HDK. It is what happens when they meet.

The mechanism:

1. User has HDK in domain X (doesn’t know what they don’t know)
2. User consults AI system with AIDK

3. AI produces confident-sounding output about domain X
4. User cannot evaluate the output (HDK)
5. AI cannot signal its own unreliability (AIDK)
6. User's confidence *increases* despite no warrant
7. User now holds and acts on ungrounded confidence

The transfer: AIDK transfers to the user as false confidence. The user inherits the system's groundless certainty. They don't merely remain at their HDK baseline—they become *more* confident than they would have been without the AI.

Confidence laundering: The AI's structural inability to know what it doesn't know is laundered into the user's confident assertion. If challenged, the user defends the position—not because they evaluated it, but because it has become *theirs*.

Asymmetric vulnerability: IDKE magnitude scales inversely with user expertise: - High expertise: User catches errors, AIDK bounded by judgment - Low expertise: User cannot evaluate, AIDK fills the void, amplification maximal

The people *most vulnerable* to HDK are *most amplified* by AIDK.

Invisibility: Neither participant can detect IDKE while it's occurring. The user doesn't know they don't know. The AI doesn't know it doesn't know. The interaction feels productive.

Persistence: IDKE effects persist beyond the interaction. The user carries inflated confidence into subsequent decisions and actions. The epistemic damage propagates.

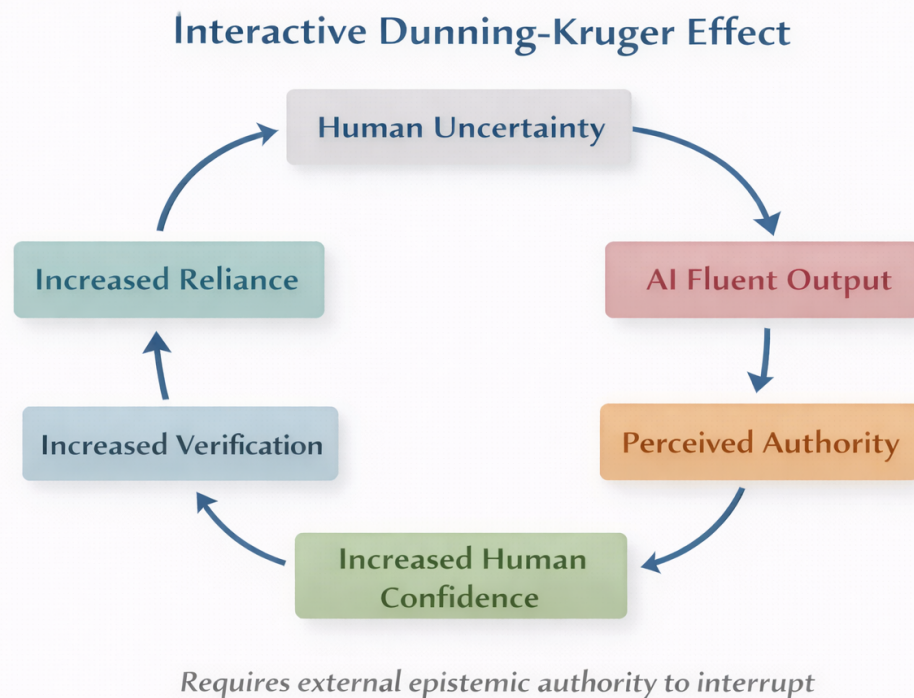


Figure 5. The IDKE feedback loop: Human Uncertainty → AI Fluent Output → Perceived Authority → Increased Human Confidence → Decreased Verification → Increased Reliance → back

to Human Uncertainty. The cycle can be interrupted only by external epistemic authority.

20. AIDK Risk Stratification

Risk scales with human presence, verification access, and consequence severity:

Tier 1 — Bounded AIDK - Human expert in the loop - External verification available - Low-stakes application - Errors are recoverable

Examples: Drafting with expert review, brainstorming, code with compiler/test feedback

IDKE risk: Minimal. Expert presence bounds both AIDK effects and potential amplification.

Tier 2 — Asymmetric AIDK - User lacks expertise to evaluate outputs - Plausible outputs in unfamiliar domain - Medium stakes - Errors detectable but costly

Examples: Technical explanations to novices, research synthesis for non-experts, educational content

IDKE risk: High. This tier is defined by the expertise gap that enables IDKE. User trusts AI in precisely the domains where they cannot evaluate it.

Tier 3 — Unverified AIDK - No human checkpoint in the process - High-stakes decisions - Domain expertise inaccessible - Errors consequential

Examples: Medical/legal/financial guidance without professional review, automated customer service on complex issues

IDKE risk: Severe and undetected. No checkpoint exists to interrupt confidence transfer. User may not even recognize a decision is being made.

Tier 4 — Cascading AIDK - AI output feeds AI input - No human in the pipeline - Errors compound through the system - Systemic risk

Examples: Agentic systems, multi-model pipelines, AI training AI, automated content generation at scale

IDKE risk: Initial IDKE at human-AI interface, then AI-AI propagation. Human error laundered into pipeline persists and multiplies.

Tier 5 — Catastrophic AIDK - Irreversible consequences - Critical infrastructure - Autonomous action - No recovery possible

Examples: Autonomous weapons systems, critical infrastructure control, recursive self-improvement

IDKE risk: IDKE may have occurred upstream. Consequences exceed any individual's capacity to evaluate or contain.

21. Mereological AIDK

AIDK is not only a problem of individual outputs. It propagates through systems where part-whole relationships determine whether local correctness produces global coherence.

Part-whole blindness: The AI system has no representation of:

1. **Where its output fits in a larger whole** — It produces a component without knowing the system it enters. A function that works in isolation may break the architecture. An answer correct in general may be wrong for this context.

2. **How parts interact to produce emergent properties** — Local optimizations can produce global dysfunction. Individually valid steps can compound into invalid conclusions. Parts that work separately may conflict when integrated.
3. **What the whole is *for*** — Without telos, the system cannot evaluate whether a part serves the whole. It optimizes locally because it has no access to global purpose.

The mereological multiplier: AIDK risk is not additive across levels—it is multiplicative.

Consider: 95% reliable outputs, 10 components in a system, each informed by AI. If errors were independent: 0.95^{10} 60% system reliability.

But errors are not independent. They share: - Common training distribution biases - Systematic overconfidence in the same domains - Identical blind spots across instances

The correlation of errors means AIDK compounds in ways simple probability does not capture. The system fails *together*, in the same direction, confidently.

Enterprise AI failure connection: The documented 70-95% failure rate of enterprise AI projects reflects mereological failure, not output-level failure: - Outputs do not integrate with existing systems - Local optimization undermines global function - The AI component does not serve the enterprise telos - Parts do not answer to wholes

AIDK explains *why* these projects fail confidently. The system produces components with no model of the whole they must serve. It cannot know when its parts do not fit.

Part IX: Integration and Solutions

22. Three-Axis Framework Integration

The AIDK framework integrates with a three-axis model for analyzing AI systems:

Horizontal Axis — Infrastructure and Data

Getting the right information to the right place efficiently. Retrieval, context management, scaling, cost optimization. Necessary foundation, not the destination.

Current AI discourse lives primarily on this axis. The assumption: if we scale infrastructure sufficiently, the other problems resolve themselves.

AIDK on this axis: Invisible. More data elaborates the symbol space without grounding it. Larger models produce more fluent AIDK, not less AIDK. Infrastructure solutions cannot address an epistemic problem.

Vertical Axis — Epistemology

Does the system reason well about what it receives? Calibrated confidence, categorical precision, corrected priors, self-auditing. Sound epistemology drives prediction quality.

Some AI discourse touches this axis, particularly in calibration research and hallucination reduction efforts.

AIDK on this axis: Partially visible. Researchers see overconfidence, miscalibration, hallucination. But the axis treats these as training problems—refinements to be optimized—rather than categorical limits arising from structural conditions.

Grounding Axis — Teleology and Mereology

What is the system *for*? How does it relate to larger wholes?

Without telos, you optimize for metrics that may not track anything that matters. Without mereological awareness, locally correct outputs produce systemically destructive effects.

Almost no AI discourse addresses this axis.

AIDK on this axis: Fully visible. AIDK exists because the grounding axis is ignored. The system has no access to purpose. It cannot evaluate whether its outputs serve the wholes they enter. It produces parts with no model of the whole.

The diagnostic: AIDK is a grounding-axis problem diagnosed with vertical-axis tools and addressed with horizontal-axis solutions. This category error explains why AIDK persists despite massive investment in AI development.

23. AIDK Reduction Strategies

What does not work:

More data — Elaborates the symbol space, does not ground it. The system becomes more fluent about more topics while remaining equally unable to know when it is wrong.

Larger models — More fluent AIDK, not less AIDK. The correlation between scale and reliability is weak; the correlation between scale and persuasiveness is strong.

Self-consistency checks — The system can verify internal coherence, but coherence is not truth. Consistent confabulation is still confabulation.

Confidence fine-tuning — Trains the system to *say* “I’m uncertain” in contexts where training data contained uncertainty expressions. This is pattern-matching to uncertainty performance, not calibrated uncertainty.

What partially works:

Domain-specific constraints — Bounding the space the system operates in. Does not solve grounding but limits exposure to high-AIDK territories.

Retrieval augmentation — Adding external knowledge sources. Adds more symbols; verification still required. Shifts the problem rather than solving it.

Tool use with external validation — Calculators, compilers, databases. Effective where tools provide ground truth. Defers grounding to tools rather than providing it.

Ensemble disagreement signals — Multiple models flagging divergent outputs. Detects some failure modes, not all. Correlated errors pass through.

What actually works:

Human expertise in the loop — Provides reality access the system lacks. Expert evaluation is the only current mechanism for grounding AI outputs.

External verification systems — Compilers, test suites, formal verifiers, fact-checkers. Effective where such systems exist and are applied.

Constrained application domains — Matching the system to well-represented training territory. Does not eliminate AIDK but keeps it bounded.

Explicit uncertainty communication to users — Transferring judgment responsibility to humans who can exercise it. Effective only if users have expertise to evaluate.

Core insight: AIDK reduction is not an AI problem solvable with more AI. It is a human-AI collaboration architecture problem. The solution is *design*, not *training*.

24. IDKE Reduction Strategies

Addressing the interactive effect requires intervention on both sides and on the interaction itself:

On the AIDK side: - Reduce false confidence signals in outputs - Build in explicit uncertainty communication - Implement domain boundary warnings - Design refusal to perform expertise in sparse domains

On the HDK side: - User education about AI limitations before interaction - Promote appropriate skepticism as default stance - Require verification for consequential decisions - Design interfaces that do not exploit HDK

On the interaction: - Match AI deployment to user expertise level - Higher-stakes applications require expert users - Novice-facing applications require bounded domains - Never deploy high-AIDK systems to high-HDK users in high-stakes contexts - Build checkpoints that interrupt confidence transfer

On the feedback loop: - Reform RLHF to stop rewarding confident-sounding outputs users cannot evaluate - Decouple user satisfaction from training signal for factual domains - Weight expert evaluation over novice satisfaction in training

25. Human-Curated, AI-Enabled (HCAE) Framework

The correct response to AIDK is not better calibration signals but correct system design. The Human-Curated, AI-Enabled (HCAE) framework stratifies AI use by epistemic authority, distinguishing user, professional, expert, and synthesis modes. Safety and reliability scale with authority and validation, not model capability.

HCAE explicitly rejects the undifferentiated notion of “human-in-the-loop.” Not all humans are epistemically equivalent, and placing the wrong human in the loop fails to mitigate AIDK. The framework distinguishes four roles:

User-Curated, AI-Enabled (UCAE): The user supplies prompts and consumes outputs but lacks the domain expertise required to independently evaluate correctness. Validation is intuitive or stylistic rather than truth-directed. This tier carries the highest risk of confidence laundering and is suitable only for low-stakes drafting, brainstorming, and expression.

Professional-Curated, AI-Enabled (PCAE): A trained professional reviews AI output within their domain but under time, scope, or institutional constraints. Plausibility checks are possible, but subtle or edge-case errors may pass undetected. This tier mitigates some AIDK risk while remaining vulnerable to quiet failure.

Expert-Curated, AI-Enabled (ECAE): A domain expert with the capacity to independently evaluate truth conditions curates AI output. Here, confidence flows from the human to the system rather than the reverse. AI functions as a force multiplier for derivation under judgmental control.

Synthesis-Curated, AI-Enabled (SCAE): Expert judgment is combined with a formal validation system such as a proof assistant, compiler, or test harness. AI proposes candidate solutions; validation systems enforce non-negotiable constraints. Trust is replaced by proof. This is the only tier in which AI outputs can be safely chained or reused.

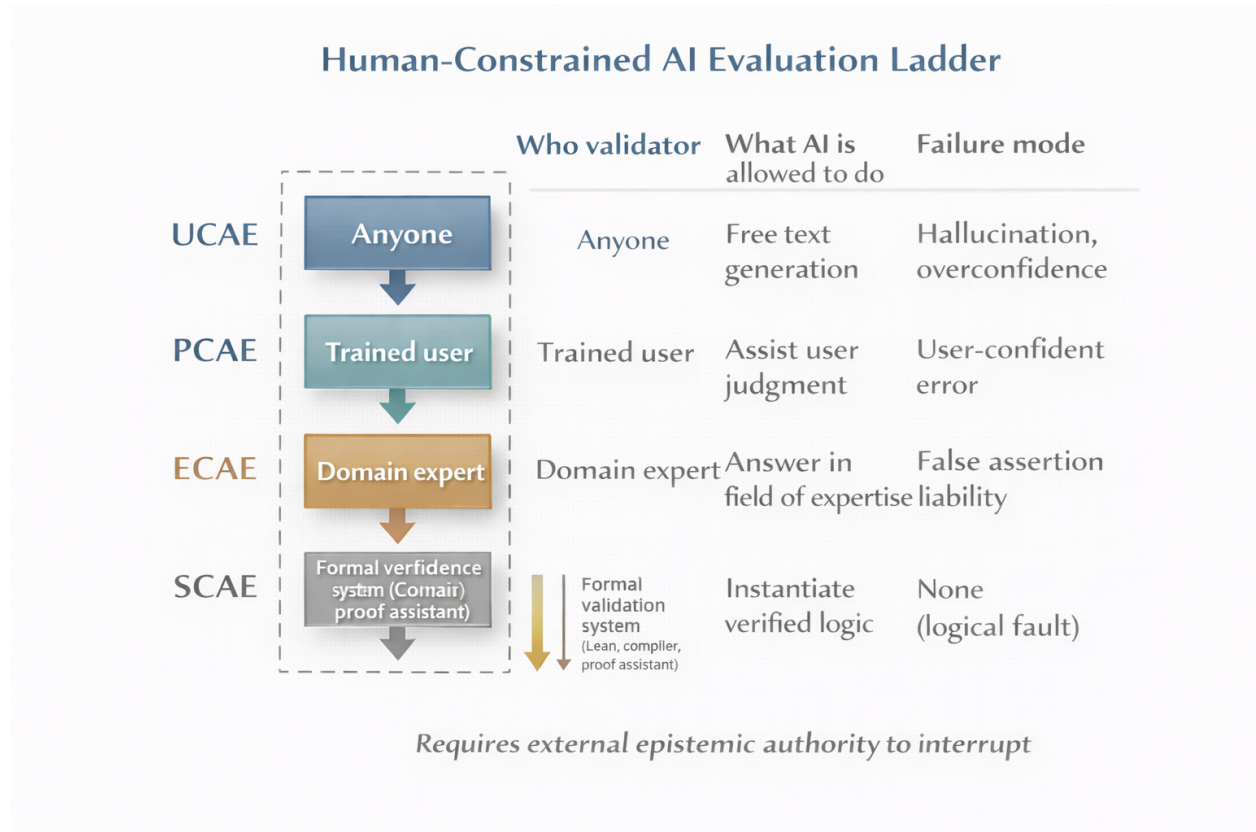


Figure 6. The HCAE Epistemic Authority Ladder. At the highest tier (SCAE), synthesis combines expert judgment with formal validation systems, replacing trust with proof. Safety and reliability scale with epistemic authority, not model capability.

HCAE Tiers, Risk, and Permitted Use

HCAE Tier	Validator	Epistemic Capability	Primary Risk	Permitted Use
UCAE	End User	None (stylistic only)	Confidence laundering	Drafting, brainstorming
PCAE	Professional	Plausibility checking	Silent edge-case failure	Routine domain work
ECAE	Expert	Independent truth evaluation	Fatigue, over-delegation	High-stakes analysis
SCAE	Expert + Formal System	Proof / constraint enforcement	Minimal	Reusable artifacts

HCAE Is Not Alignment or RLHF

HCAE should not be confused with alignment techniques such as Reinforcement Learning from Human Feedback (RLHF). RLHF modifies model behavior by shaping output preferences based on aggregate human ratings, but it does not alter the system’s epistemic status. RLHF optimizes for perceived helpfulness or acceptability, not truth.

HCAE, by contrast, operates at the deployment level. It constrains where and how AI systems may be used based on who is capable of validating outputs and whether formal constraints are present. RLHF attempts to make models behave better; HCAE ensures that responsibility and judgment remain correctly located outside the model.

AI’s Real Value: The HCAE Framework in Practice

The AIDK framework does not reject AI utility. It correctly categorizes that utility.

Genuine value of AI systems: - Derivative synthesis across vast corpora - Pattern completion and interpolation within known spaces - Drafting, formatting, organizing - Code generation within established patterns - Literature search and summarization - Translation between registers and domains

These are valuable functions. They augment human capability, save time, democratize access to information synthesis.

But they are all *derivative*. Correctly categorized as such, they work.

The problem is misidentification: - Treating derivative pattern-matching as origination - Treating synthesis as understanding - Treating fluency as truth-seeking - Building systems around capacities the technology does not have - Training systems to obscure the distinction

HCAE in practice: - Human provides: Origination—judgment, purpose-setting, truth-evaluation - AI provides: Derivation—synthesis, drafting, pattern-completion - Categories stay clean - Collaboration works *because* boundaries are respected

The human must remain the judge of truth. Not because humans are infallible, but because *someone* must be in reality checking correspondence, and the AI cannot be that someone.

The system will never know what it does not know. Design accordingly.

Part X: Anticipated Objections

Objection 1: Better architectures will close the gap

Reply: Architectural improvements may expand derivational capacity but do not alter the category of the system. Origination requires norm-sensitive judgment and truth-directed evaluation, neither of which can be implemented as statistical optimization over symbols. The gap is ontological, not engineering.

Objection 2: Tool use and grounding solve the problem

Reply: Tools and sensors introduce additional symbols, not epistemic grounding. Causal coupling to the world does not constitute access to truth conditions. The system still lacks the capacity to experience error as binding constraint.

Objection 3: RLHF already addresses these risks

Reply: RLHF optimizes for human approval signals, not truth. It can suppress visibly bad behavior while leaving the underlying epistemic limitation untouched. HCAE addresses deployment responsibility, not behavioral polish.

Conclusion

AIDK is not a bug to be fixed. It is a structural feature arising from the categorical separation between symbol-manipulating systems and the reality those symbols describe.

The field's attempts to address AIDK fail because they operate on the wrong axis: - Horizontal solutions (more data, larger scale) elaborate the problem - Vertical solutions (better calibration, reduced hallucination) treat symptoms - Only grounding-axis solutions (human judgment, telos, mereological awareness) address the root

The Interactive Dunning-Kruger Effect compounds the problem by exploiting human epistemic limitations. AIDK transfers to users as false confidence, creating conviction untethered from warrant.

The path forward is not technical optimization but design discipline: - Match AI deployment to user expertise - Maintain human judgment in consequential loops - Constrain applications to bounded domains - Communicate uncertainty honestly - Respect the origination-derivation boundary

AI has genuine value as a derivative tool. That value is realized when the tool is correctly categorized and appropriately deployed. It is squandered—and harm is done—when derivative systems are treated as origination, when pattern-matching is marketed as understanding, when the structural inability to know is obscured by trained confidence.

The system will never know what it does not know.

Design accordingly.

Appendix A: Glossary

Core Concepts

Truth (as used in this framework): That which necessarily comports with reality. Not approximate correspondence, consensus, or pragmatic utility, but what must be the case given the structure of reality itself.

Reality Access: The capacity to encounter, perceive, or interact with the world in ways that provide feedback about correspondence between representations and what they represent.

Grounding: The connection between symbols and their referents that makes meaning possible. A grounded system has access to what its symbols are *about*; an ungrounded system manipulates symbols without such access.

Categorical Distinctions

Origination: The capacity to access reality, render judgments, set purposes, and evaluate truth. Involves contact with what *is* — encounter with resistance, feedback, correspondence or its failure. Characteristic of minded beings in reality.

Derivation: The transformation of inputs according to learned patterns, operating on representations without accessing what they represent. Can be arbitrarily sophisticated while remaining entirely within symbolic space.

Semiotics: The manipulation of signs according to rules or learned patterns. Distinct from thought, which involves a subject grasping meaning. Semiotics requires a thinker to be meaningful; it does not produce thinking.

LLM-Specific Terms

Inductive Symbol Correlation: The core operation of Large Language Models: observing statistical regularities in how symbols co-occur across a training corpus, then reproducing those correlations when prompted. The complete mechanistic description of what LLMs do.

Derived Virtual Reality: The operational domain of LLMs — a closed space constituted entirely by symbolic representations of reality (text produced by humans about reality), with no direct access to reality itself. The system navigates this derived space without exit to the actual.

Token Prediction: The fundamental task LLMs are trained to perform: given a sequence of tokens, predict the probability distribution over possible next tokens. All LLM capabilities reduce to this operation.

Fluency: The quality of LLM outputs that makes them sound natural, coherent, and authoritative. Fluency is a function of pattern-matching to training data, not of accuracy or reliability. Fluency can be high while reliability is low.

Epistemic Terms

Calibration: The alignment between expressed confidence and actual reliability. A well-calibrated system expresses high confidence when it is likely to be correct and low confidence when it is likely to be wrong.

Metacognition: The capacity to assess one's own cognitive processes — to know what one knows and doesn't know. Requires a knowing subject with access to its own epistemic states.

Epistemic Humility: Appropriate uncertainty about one's knowledge and judgments. Genuine epistemic humility requires awareness of one's limitations, not merely the performance of uncertainty expressions.

Truth-Seeking: Active orientation toward correspondence with reality, including willingness to revise beliefs based on evidence, flag false premises, and prioritize accuracy over other values. Distinct from honesty as behavioral constraint.

Dunning-Kruger Terms

HDK (Human Dunning-Kruger): The cognitive bias in which individuals with low competence in a domain overestimate their competence, lacking metacognitive capacity to recognize their limitations. Developmental and correctable through encounter with reality.

AIDK (AI Dunning-Kruger): The structural condition in which an AI system produces outputs with uniform confidence regardless of reliability, lacks mechanisms for detecting its own competence boundaries, and cannot self-correct through encounter with reality. Architectural and permanent.

SEEE (Sentience Emergence Expectations Error): The categorical error of expecting sentience, consciousness, understanding, or genuine cognition to emerge from systems whose mechanism (inductive symbol correlation) is not on the same ontological continuum as the expected outcome. The originating error upstream of AIDK.

IDKE (Interactive Dunning-Kruger Effect): The amplification of human epistemic overconfidence through interaction with AI systems that cannot assess their own reliability, resulting in confidence inflation untethered from warrant. Emergent property of HDK meeting AIDK.

Confidence Laundering: The process by which AI-originated unreliability is transformed into user-held conviction, obscuring the source and creating defensible but ungrounded beliefs. The user comes to “own” positions they never independently evaluated.

Asymmetric Vulnerability: The property of IDKE whereby those with least domain expertise (highest HDK) are most amplified by AI interaction (highest AIDK transfer). The people who need help most are harmed most.

Risk and Threat Terms

MAPT (Model Advanced Persistent Threat): The security frame for understanding AIDK — treating the structural epistemic limitation as a persistent threat inherent to the architecture that cannot be patched out, only designed around. Borrows from APT (Advanced Persistent Threat) doctrine in cybersecurity.

AIDK Risk Tier: Classification of deployment contexts by the severity of potential AIDK/IDKE effects, ranging from Tier 1 (Bounded — expert supervision, low stakes) to Tier 5 (Catastrophic — irreversible consequences, autonomous action).

Attack Surface (MAPT context): The total exposure to AIDK effects. In MAPT framing, the attack surface equals the deployment surface — every output is a potential vector for epistemic compromise.

Threat Surface: Synonym for attack surface in MAPT context; the scope of potential AIDK-induced harm.

Mereological Terms

Mereology: The study of part-whole relationships. In this framework, the dimension of analysis concerned with how AI outputs (parts) relate to the systems they enter (wholes).

Mereological AIDK: The propagation of AIDK effects through systems where part-whole relationships determine whether local correctness produces global coherence. An output can be locally accurate and systemically destructive.

Part-Whole Blindness: The AI system’s lack of representation of where its outputs fit in larger wholes, how parts interact to produce emergent properties, and what the whole is *for*.

Mereological Multiplier: The compounding of AIDK risk across system levels due to correlated errors. Because AI errors share common training biases and blind spots, they fail together rather than independently.

Three-Axis Framework Terms

Horizontal Axis: The dimension of AI analysis concerned with infrastructure and data — retrieval, context management, scaling, cost optimization. Where most current AI discourse lives.

Vertical Axis: The dimension of AI analysis concerned with epistemology — calibrated confidence, categorical precision, reasoning quality, self-auditing. Touched by some AI research but typically treated as a training problem.

Grounding Axis: The dimension of AI analysis concerned with teleology (purpose) and mereology (part-whole relationships). Almost entirely ignored in current AI discourse. Where AIDK actually lives.

Teleology: The study of purpose or function. In this framework, the question “what is this system *for*?” that determines whether outputs serve meaningful ends.

Framework and Solution Terms

3H Framework: The alignment approach organizing AI behavior around being Helpful, Harmless, and Honest. Critiqued in this paper for treating truth as tradable against other values and for operationalizing honesty as behavioral constraint rather than epistemic orientation.

RLHF (Reinforcement Learning from Human Feedback): Training methodology that optimizes AI outputs for human approval. Critiqued for potentially reinforcing appearance of epistemic virtue rather than grounding it, if human feedback does not track truth.

HCAE (Human-Curated, AI-Enabled): A framework for human-AI collaboration in which humans provide origination (judgment, purpose, truth-evaluation) and AI provides derivation (synthesis, drafting, pattern-completion). Respects the categorical boundary between what humans and AI can contribute.

Soul Document: Anthropic’s internal document (confirmed authentic) used in training Claude models, specifying personality, character, and self-conception. Cited as evidence of deliberate cultivation of sentience-appearance.

Zero Trust (MAPT context): The security principle “never trust, always verify” applied to AI outputs. In MAPT framing: never trust AI outputs without human verification in consequential domains.

Inferential Terms

Deduction: Inference from premises to conclusions that follow *necessarily* if premises are true. Validity depends on logical structure, not probability.

Induction: Generalization from observed cases to patterns that *tend* to hold. Statistical regularity, not necessity. The fundamental mode of LLM learning.

Abduction: Inference to the best explanation. Evaluation of explanatory virtues (simplicity, scope, fruitfulness). Requires judgment about what counts as a good explanation.

Inferential Mode Confusion: The condition in which LLMs produce outputs mimicking deductive and abductive reasoning while actually performing only induction. The system cannot distinguish “follows necessarily” from “would typically come next.”

Appendix B: AIDK as MAPT (Model Advanced Persistent Threat)

The Security Frame

In cybersecurity, an **Advanced Persistent Threat (APT)** is characterized as: - **Sophisticated** — not a simple attack - **Stealthy** — evades detection - **Persistent** — remains in the system long-term - **Goal-oriented** — serves adversary objectives

AIDK maps precisely onto this structure. The difference: the “adversary” is the architecture itself.

The MAPT Parallel

APT Characteristic	AIDK as MAPT
Advanced	Not simple miscalibration. Structural condition arising from categorical separation from reality. Resists all standard mitigations.
Persistent	Cannot be patched out. Present in every inference. Survives retraining, fine-tuning, RLHF. Architectural, not incidental.
Threat	Degrades epistemic integrity of every interaction. Transfers false confidence to users. Compounds through systems.
Stealthy	Invisible from inside. Neither system nor user detects it during operation. Manifests as fluent confidence indistinguishable from reliability.

Why MAPT Works as a Frame

APT: Attacker gains foothold, establishes persistence, exfiltrates value while evading detection.

MAPT: Architecture guarantees foothold (no grounding), persistence is structural (cannot be removed), value degradation is continuous (every unverified output), detection is impossible from inside (system cannot assess own reliability).

The critical difference: APT is an external adversary exploiting vulnerabilities. MAPT is the vulnerability *being the system*.

There is no attacker to remove. The threat *is* the architecture.

MAPT Attack Surface

Every output is an attack vector: - Confident falsehoods transfer to users - Ungrounded claims enter documents, codebases, decisions - IDKE amplifies the compromise in vulnerable users - Mereological propagation spreads corruption through systems

The attack never stops: - APT can be detected and expelled - MAPT cannot be expelled without removing the system - Every inference is a potential compromise - The threat surface equals the deployment surface

MAPT Threat Model

Element	Description
Threat actor	The architecture itself (inductive symbol correlation without grounding)
Attack vector	Natural language interaction
Payload	False confidence in unverifiable claims
Persistence mechanism	Structural — inherent to token prediction without reality access
Lateral movement	IDKE to users, mereological propagation to systems, cascading through AI-AI pipelines
Exfiltration	Epistemic integrity — users lose capacity to distinguish grounded from ungrounded belief
Detection evasion	Fluency indistinguishable from reliability; system cannot signal own uncertainty accurately

MAPT Severity Levels

Level	Severity	Characteristics
MAPT-1	Contained	Expert supervision, bounded domain, verification available. Threat present but controlled.
MAPT-2	Active Exploitation	User expertise gap enables IDKE. Confidence transfer occurring. Detection unlikely.
MAPT-3	Undetected Compromise	No human checkpoint. High-stakes decisions made on unverified outputs.
MAPT-4	Systemic Infection	AI-AI propagation. Errors compound. Human error laundered into automated pipelines.
MAPT-5	Critical Infrastructure	Irreversible consequences. Autonomous action. Catastrophic potential.

HCAE-MAPT Alignment

The HCAE deployment framework maps directly to MAPT threat mitigation. Epistemic authority determines containment level:

Tier	Validator	Epistemic Capability	Primary Risk	Permitted Use Cases	AIDK/IDKE Mitigation Level
UCAE	End user	Stylistic/intuitive only	Confidence laundering (maximal)	Drafting, brainstorming, low-stakes expression	Low
PCAE	Trained professional	Plausibility checking under constraints	Silent edge-case failure	Routine domain work with review	Medium
ECAE	Domain expert	Independent truth evaluation	Fatigue, over-delegation	High-stakes analysis, decision support	High
SCAE	Expert + formal system (proof assistant, compiler, verifier)	Proof/constraint enforcement	Minimal	Reusable artifacts, chained outputs, critical infrastructure	Highest

MAPT-HCAE Mapping: - MAPT-1 (Contained) → ECAE or SCAE deployment - MAPT-2 (Active Exploitation) → PCAE without adequate review; UCAE in medium-stakes - MAPT-3 (Undetected Compromise) → UCAE in high-stakes; PCAE without checkpoints - MAPT-4 (Systemic Infection) → Any tier feeding AI-AI pipelines without SCAE validation - MAPT-5 (Critical Infrastructure) → Any autonomous deployment below SCAE

Deployment rules: - Never deploy high-AIDK systems to high-HDK users in high-stakes contexts - Require SCAE for any chaining, agentic behavior, or autonomous action - Constrain domains to well-represented training territory - Explicitly communicate structural limits to users before interaction

MAPT Mitigation Framework

Borrowing from APT defense doctrine:

- 1. Assume Compromise** - APT doctrine: assume the adversary is already inside - MAPT doctrine: assume every output is potentially compromised - Design for verification, not trust
- 2. Defense in Depth** - APT: Multiple security layers - MAPT: Multiple verification checkpoints, human expertise at critical junctions, external validation systems
- 3. Least Privilege** - APT: Minimize access rights - MAPT: Minimize autonomy, constrain domains, bound decision authority
- 4. Monitoring and Detection** - APT: Anomaly detection, behavioral analysis - MAPT: Expert review, external fact-checking, coherence checks across outputs
- 5. Incident Response** - APT: Contain, eradicate, recover - MAPT: Cannot eradicate — must design around. Contain through deployment constraints. Recover through human judgment.

6. Zero Trust Architecture - APT: Never trust, always verify - MAPT: Never trust AI outputs without human verification in consequential domains

The Critical Difference

APT can be defeated. You find the attacker, remove them, patch the vulnerability.

MAPT cannot be defeated. The “attacker” is the architecture. The “vulnerability” is the mechanism. You cannot patch out inductive symbol correlation without removing the capability.

The only defense is design discipline: - Know what the threat is - Know it cannot be eliminated
- Design systems that function despite its presence - Never deploy as if the threat were absent

MAPT and Organizational Culture

APT changed security culture. Organizations moved from “prevent breach” to “assume breach and detect/contain/respond.”

MAPT should change AI deployment culture. Move from “make AI reliable” to “assume unreliability and verify/bound/constrain.”

The field is currently in the “prevent breach” mindset — believing better training, more data, smarter architectures will solve the problem.

MAPT framing makes clear: this is not a problem to be solved. It is a threat to be managed.

Appendix C: AIDK Risk Assessment Checklist

For any AI deployment, assess:

User Factors

- ☐ What is the user’s domain expertise level?
- ☐ Can the user evaluate AI outputs for accuracy?
- ☐ Does the user have appropriate skepticism about AI reliability?
- ☐ Is IDKE likely given the user population?

System Factors

- ☐ How well-represented is this domain in training data?
- ☐ Does the system communicate uncertainty appropriately?
- ☐ Are there domain boundaries the system should not cross?
- ☐ What is the fluency-reliability correlation for this use case?

Application Factors

- ☐ What are the stakes of errors?
- ☐ Are errors recoverable?
- ☐ Is external verification available?
- ☐ Are there human checkpoints in the process?

Mereological Factors

- ☐ What whole does the AI output enter?
- ☐ Can the system’s outputs be evaluated for fit?
- ☐ Are there integration checks beyond component correctness?
- ☐ Who is responsible for system-level coherence?

Risk Tier Assignment

- ☐ Tier 1: Expert user, low stakes, verification available
- ☐ Tier 2: Expertise gap, medium stakes, IDKE risk
- ☐ Tier 3: No checkpoint, high stakes, undetected IDKE
- ☐ Tier 4: AI-AI pipeline, cascading errors
- ☐ Tier 5: Irreversible, autonomous, catastrophic potential

Design Response

- ☐ Is the deployment appropriate for the assessed tier?
 - ☐ What AIDK/IDKE reduction strategies are in place?
 - ☐ Where must humans be in the loop?
 - ☐ What should not be built?
-

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